Workshop Task

Creating and Hosting a Website

These instructions were written by Dr. Joanne Armitage and Dr. Chris Birchall.

Context

In this workshop you will begin to create content and host it online. It will familiarize you with the Digital Media development environment, including some of the various production software available, and some services for hosting content online. You should use this workshop to create an online portfolio which can be developed continuously during your time as a Digital Media student, and use the associated resources provided to further your web design and development skills.

In this workshop you will:

- Activate your University of Leeds CampusPress website and familiarise yourself with it;
- Create a HTML file using your choice of editing software;
- Create (or access an existing) personal account on GitHub and upload your website to it so that it can be viewed publicly;
- Begin to think about the structure of an online portfolio.

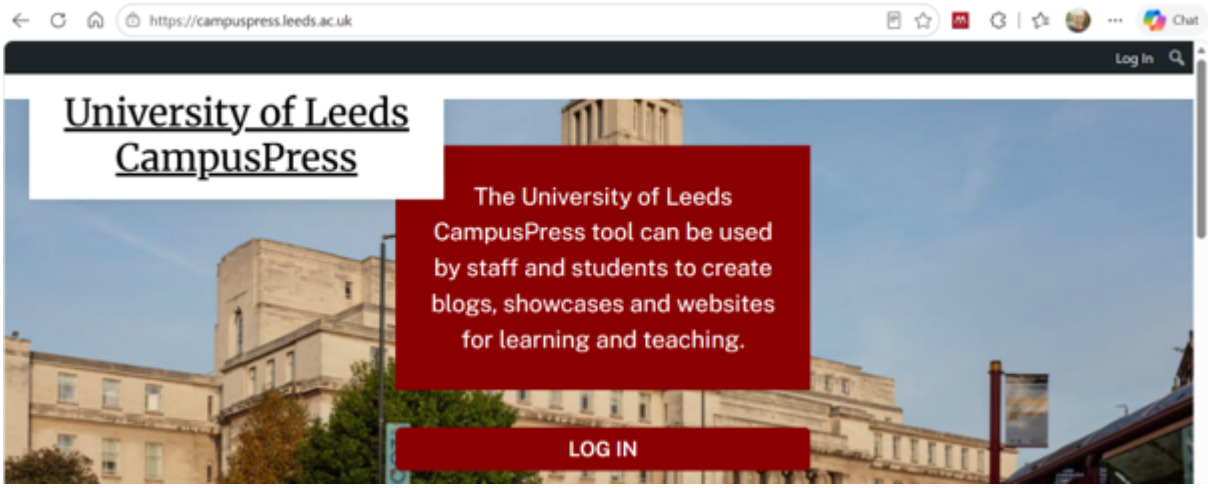
The workshop is very basic, assuming no prior expertise in HTML/CSS or the various tools used. If you have existing web design skills, take the opportunity to further develop the initial portfolio page as you see fit.

We are going to look at three distinct practices:

1. Using a content management system that builds a website for you – in our case CampusPress
2. Build a web page using code – in our case HTML and CSS
3. Store our web page code online to make it available publicly as a website – in our case using GitHub Pages

Refer to our Terms and Conditions (https://leeds365-my.sharepoint.com/:b:/g/personal/fbsctb_leeds_ac_uk/IQARbF9s89HkSoM8HE8RA0NwAaOLbH91-_P827Q5PAPkPQs?e=U53QEh) ***document for details of how you are expected to use these systems (and what not to do!) and make sure you back up work on your University of Leeds OneDrive storage where possible.***

Activating and Managing Your University CampusPress Site



CampusPress is a tool for creating blogs and websites for learning purposes. Powered by WordPress, you can create and manage your own blogs, using a range of accessible themes. You can create traditional websites, blog formats, or a variety of other styles of online platform using the selection of themes available. You can also have more than one website hosted here.

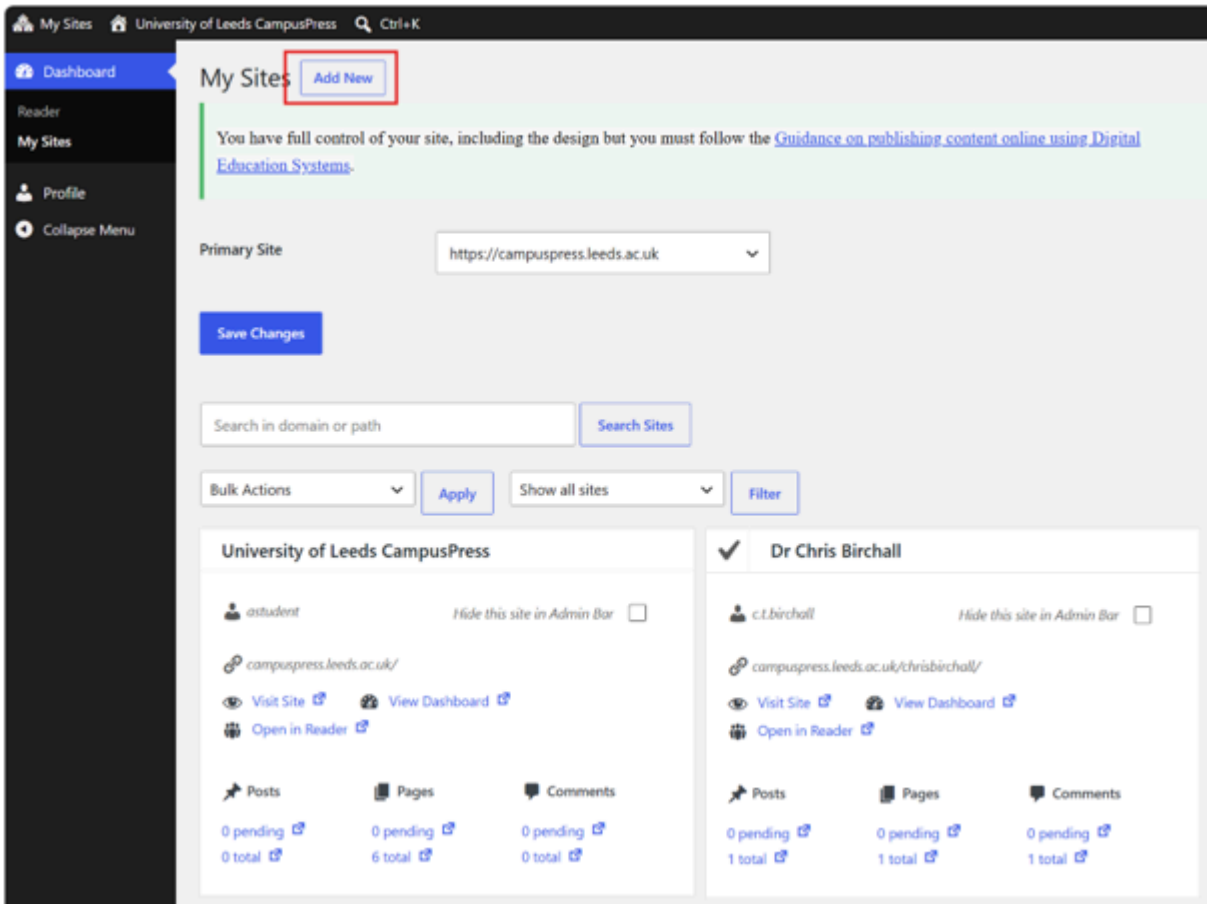
For this module, you will create a portfolio of work to showcase what you have done during your studies. You may not build all of your web content within CampusPress itself as it is somewhat restrictive, but you can take screen shots and showcase it here, or include links from reflective blog posts to allow visitors to access the work. Beyond this you could create a website that serves as a base for your group project practical work. You could experiment with interactive and community features, or see what other features are available through the plugin library.

See here for full details: https://desystemshelp.leeds.ac.uk/campuspress-students/ (https://desystemshelp.leeds.ac.uk/campuspress-students/)

Setting up your CampusPress site

To set up your site, visit the link above and click login, then sign in with your University details. Once setup completes you will be shown the Dashboard and any default websites that you are part of will be shown in white boxes at the bottom of the screen — there may be none at first!

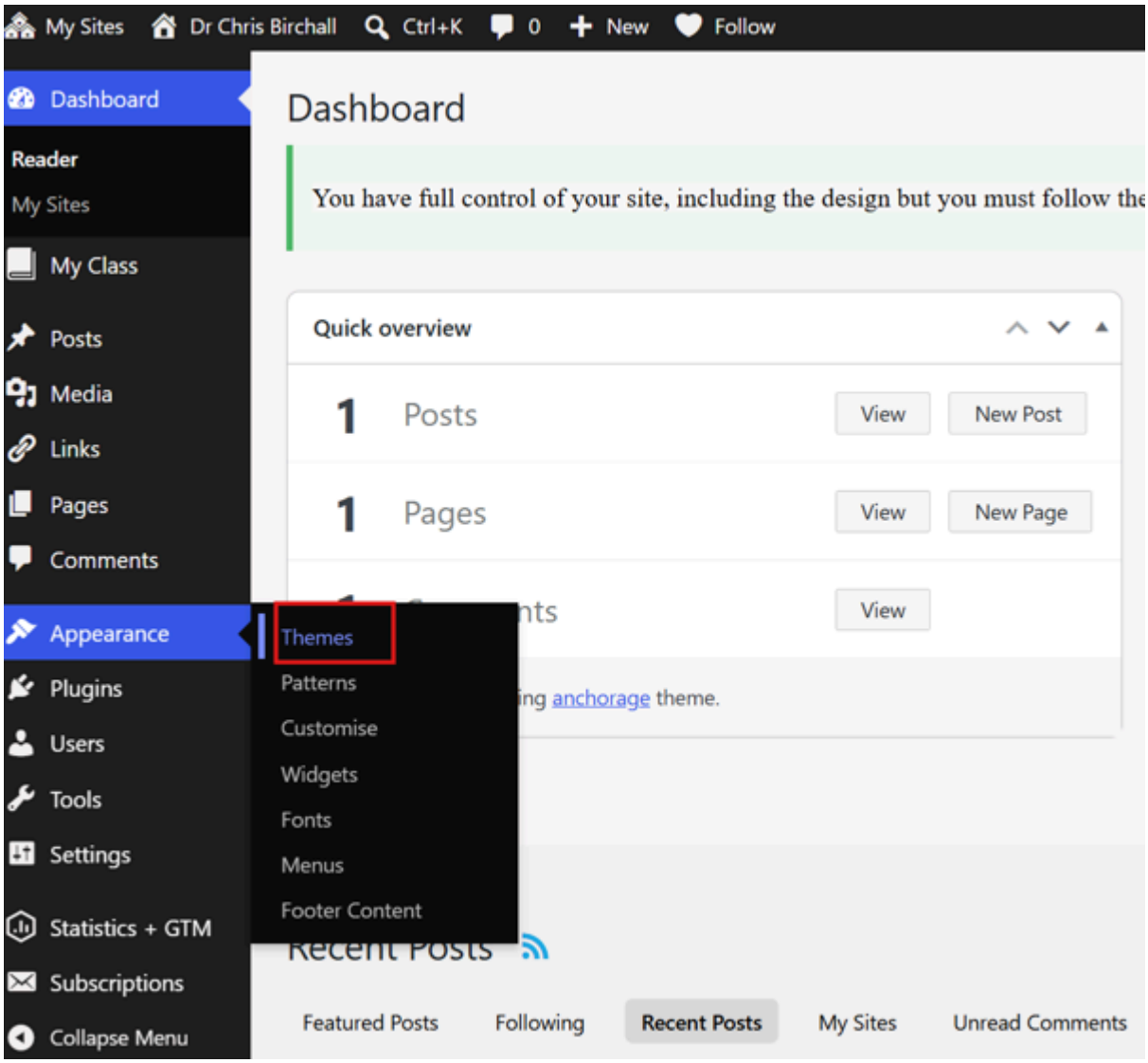
To create your first website just click on the Add New button at the top of the Dashboard



You can then choose a name and title and select a template for your site.

- In the first box, put something short – ideally one word with no spaces; this will be the end part of the URL of your site (e.g. https://campuspress.leeds.ac.uk/(https://campuspress.leeds.ac.uk/chrisbirchall)**chrisbirchall (https://campuspress.leeds.ac.uk/chrisbirchall)**) for mine.
- In the second box put a really short descriptive title (such as "Chris Birchall's portfolio").

Don't worry about a theme yet — there won't be many to choose from at this point and you can change it later. Click **Create Site**. You will be directed back to the Dashboard. You will see your site in a box at the bottom of the page. Within the box you can click on **View Dashboard.** This Dashboard will let you change the design and content of your website. First, have a look at the available **themes**:



These themes control the structure, design and aesthetics of your website. You can choose any of the options and you can edit them too.

Next create content with **Posts** or **Pages**, and explore the available **Plugins**.

There is a lot to learn about CampusPress (or WordPress which this is based upon). One resource to help is this **CampusPress Knowledgebase site**: https://docs.campuspress.com/en (https://docs.campuspress.com/en)

Building a Website using Code

Content Management Systems like CampusPress control and limit what you build. Sometimes you will want to be more creative and innovative and original. In those cases, you need to get stuck into writing or utilising code.

Let's take this step by step.

Creating a Sample Homepage

On your laptop:

- Create a folder called 'mywebsite' (this is where you are going to store ALL of your web work).
- Open an editor of your choice (VS Code, TextEdit, Notepad++, etc.) and create a new HTML file.
- You can download VS Code here (https://code.visualstudio.com/download?_exp_download=fb315fc982).
- Save the file as 'index.html', in your 'mywebsite' folder.
- Edit the file to contain some content (maybe your name).

Basic HTML Code:

```
<!doctype html>
<html>
<head>
<title> put your document's title here: this appears at the top of the browser window and is returned when users do a search </title>
</head>
<body> put your document's main content here </body>
</html>
```

Now add some content to the 'index.html' file that you created. Depending on your choice of editing software, your file may already have some skeleton HTML, such as a doctype definition. If not, add HTML head and body sections. Now add:

- a title
- some simple paragraphs
- a sensible heading hierarchy — try to use at least two levels of heading
- any of the other tags you have learnt

If you don't know what to write, try this:

- Your name as the main header
- A sentence or two introducing yourself
- The modules you are studying, under a subheader, as a list
- Some further information about you (hobbies, interests, work experience, new media experience), also under a subheader

See the sample code below for examples of how to mark up content with HTML. There are also a list of great online resources that can teach you all about web design and development in the Workshop Preparation & Reading folder and towards the bottom of this page.

Adding Some Style to the Page

Add a link to an external stylesheet in the <head> section of your HTML, as below:

```
<link href="your-stylesheet-name.css" rel="stylesheet" type="text/css" />
```

Now create the stylesheet. This is just a new file, containing your style descriptors (see below) and saved with the file name specified in the “href” part of the <link> tag above.

Create styles for some of the selectors listed below.

Save the stylesheet within the same folder as your HTML page, above, and upload it to the server. Your stylesheet can be saved with any name (as long as the file extension is .css), but the name of the file must match that specified in the “href” of the “link” element in the head section of your HTML page.

Once you've saved your stylesheet, retest your page in the browser to see how it looks.

Publishing your Website using GitHub Pages

Now, how can we get others access to your site?

There are lots of ways to publish content online, through content management systems, web hosting services, cloud storage and many other routes. A flexible way to publish content online is GitHub. GitHub is a widely used platform for storing, managing, and sharing digital files and code. It is commonly used by web developers, software engineers, designers, digital media practitioners, and researchers to organise project work, keep track of changes, and collaborate with others.

Using GitHub can help you develop professional digital working practices. By storing your files in a repository, you can maintain a record of your development process, experiment with different versions of your work, and back up important project materials. GitHub also provides tools for documenting your project, allowing you to explain your goals, design decisions, and development process through README files and other project documentation. There are quite a few different software products associated with GitHub as well as the GitHub version of Copilot.

For you, when creating websites or other web–based projects there is **GitHub Pages**, a free hosting service provided through GitHub. GitHub Pages allows you to publish HTML, CSS, and JavaScript projects directly to the web, making it easy to share prototypes, portfolios, and project outputs with various audiences. A GitHub Pages site can be updated throughout development and provides a convenient way to demonstrate your work online. As you can store other resources such as spreadsheets, images and other files in GitHub you can create whole web applications there, using JavaScript to link to any cloud services that you want to utilise.

Creating an account, a repository, and enabling Pages

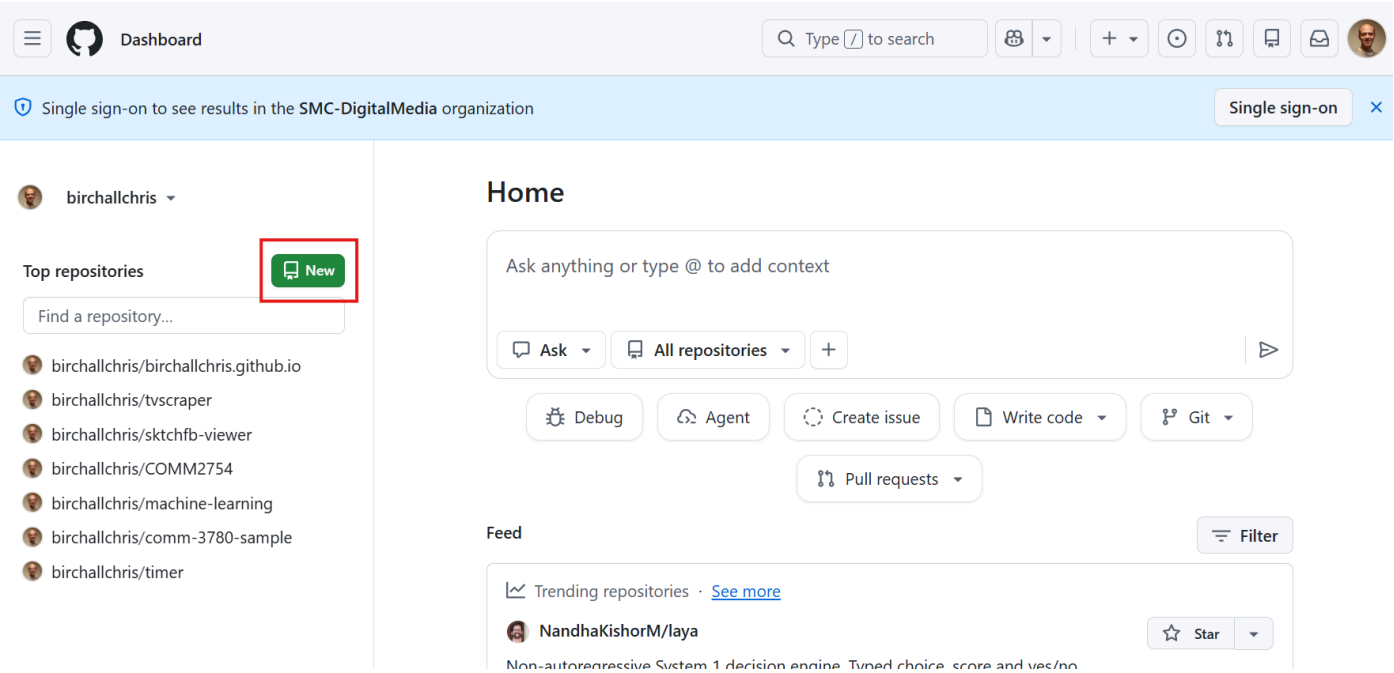
To create a GitHub account, simply go to https://github.com/ (https://github.com/) and sign up with an email address. *This can be any email address and need not be your University email address (using a personal account means you can keep it after graduation)*. This will be your own

personal GitHub account for you to use for anything, for life. It is not a University system. Verify your email address immediately using the link in the GitHub email.

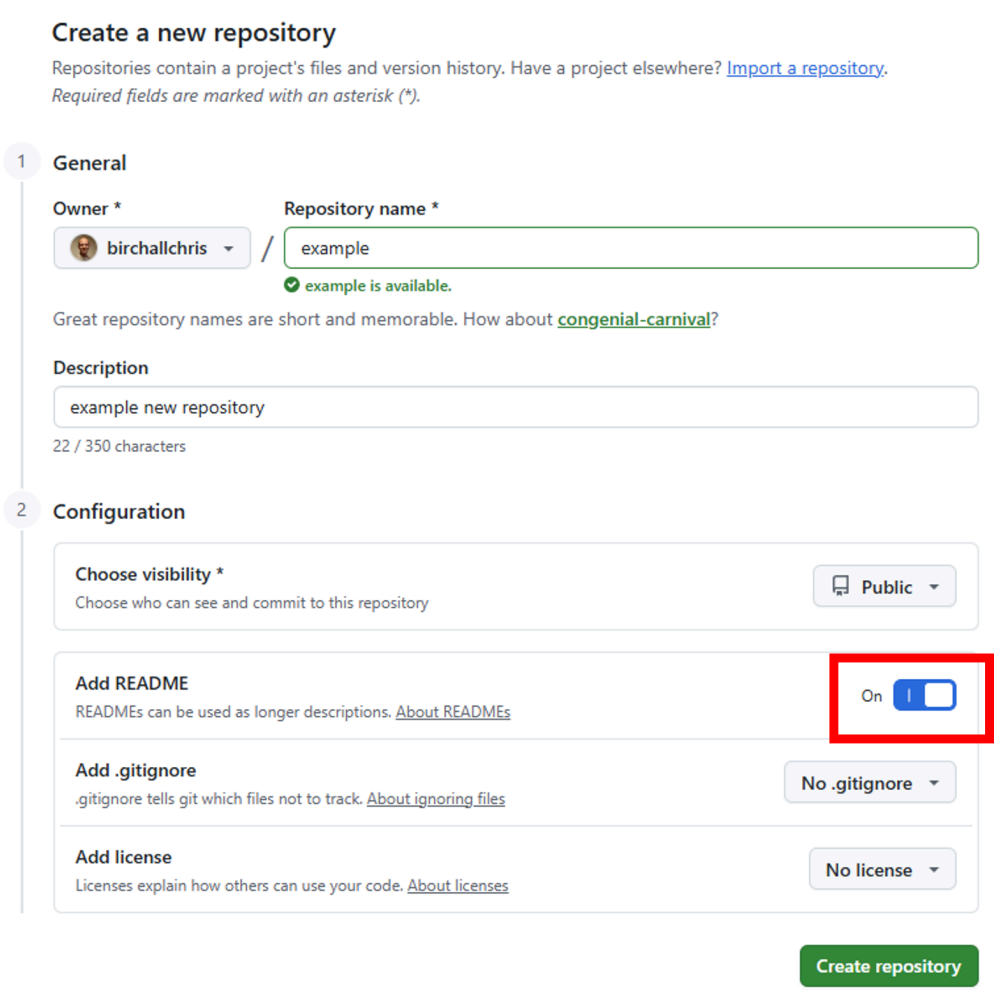
If you use a VPN or other security software, it can make GitHub suspicious and they might auto-block your account. It is maybe best to turn off your VPN when creating the account.

If you already have a GitHub account you can use that, or create a new one, your choice.

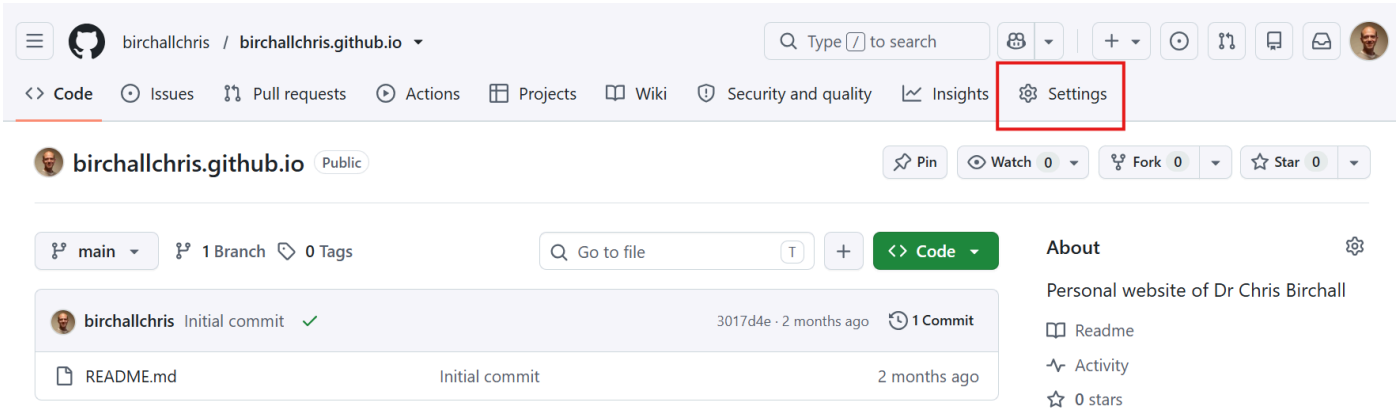
Once signed in, you will see your homepage and you'll be able to create a repository using the 'New' button [Note from Holly: if the 'New' button is not showing as pictured below, you can click the + icon in the top right-hand side, and select 'New repository']:



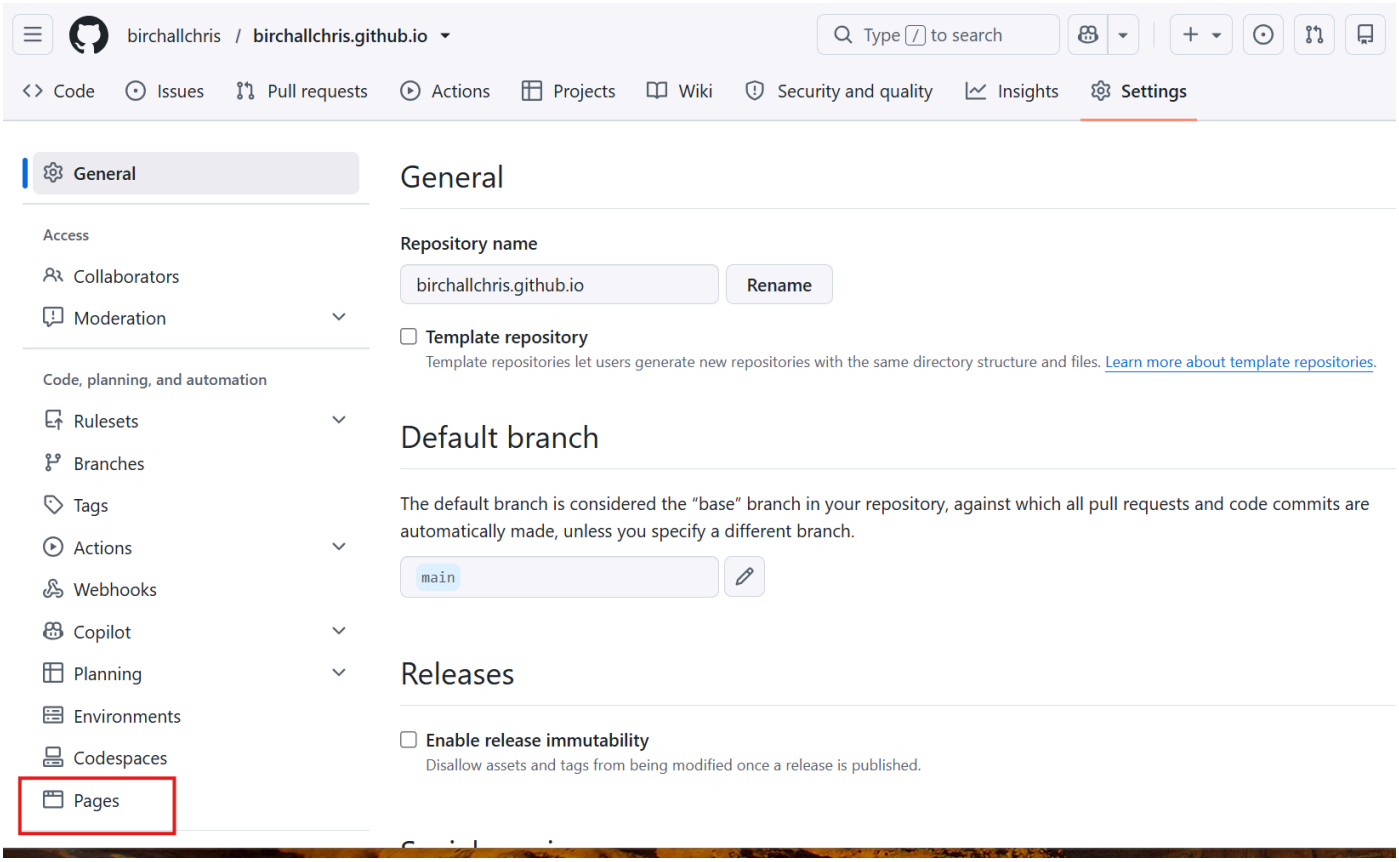
A repository is a space to store project files. You can configure a repository to be public or private or shared, to have collaborators or to be used in different ways. **Create** a repository for COMM5780. When you do this, be sure to add the **Readme file option**, as it makes life easier and provides a default website which is visible to viewers until you upload more files:



Click **Create Repository**, then, enter the repository and click on the **Settings** link in the menu:



In the settings menu, click on the **Pages** item – this will allow you to set up the Pages feature, which turns your repository into a website.

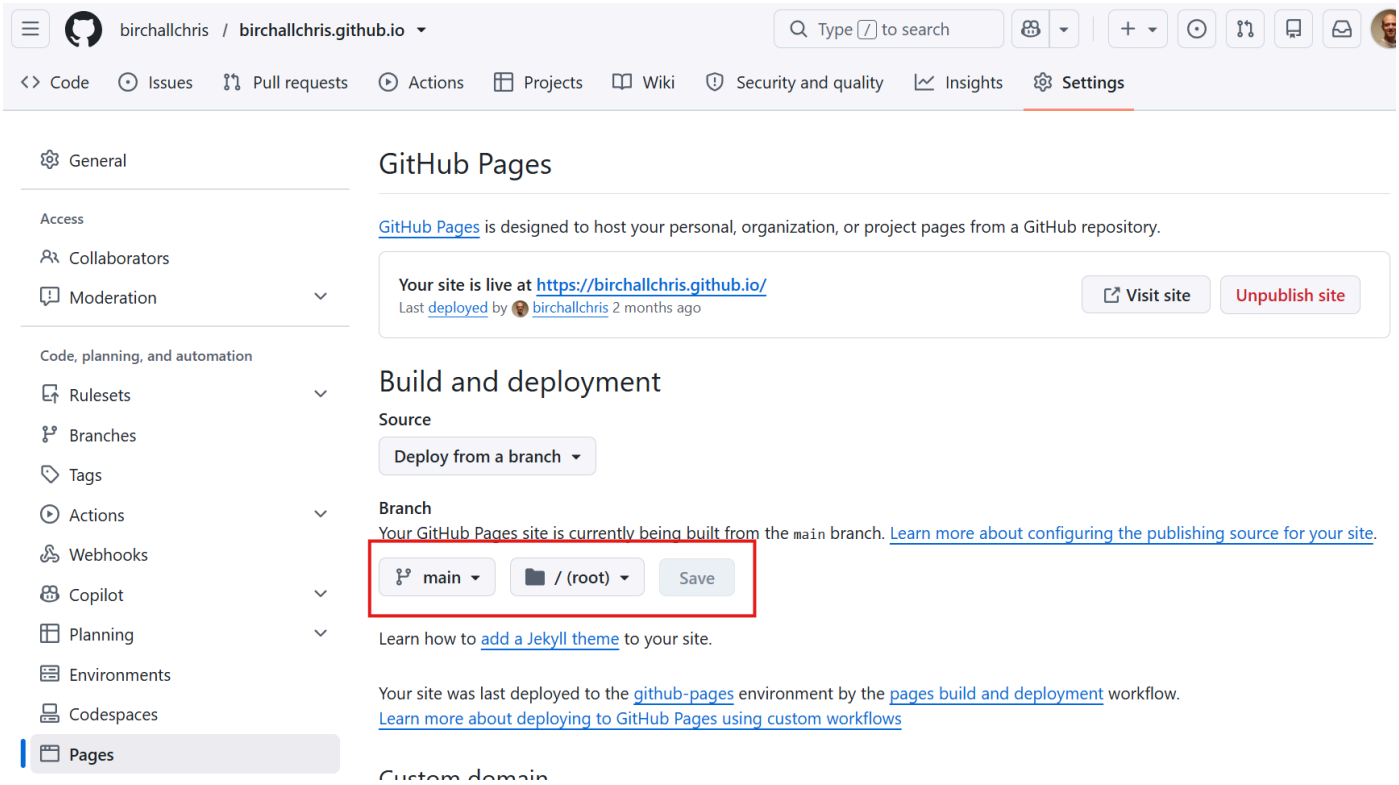


In the Pages settings, under **Build and deployment**, set:

- **Source:** deploy from branch
- **Branch:** main
- **Folder:** / (root)

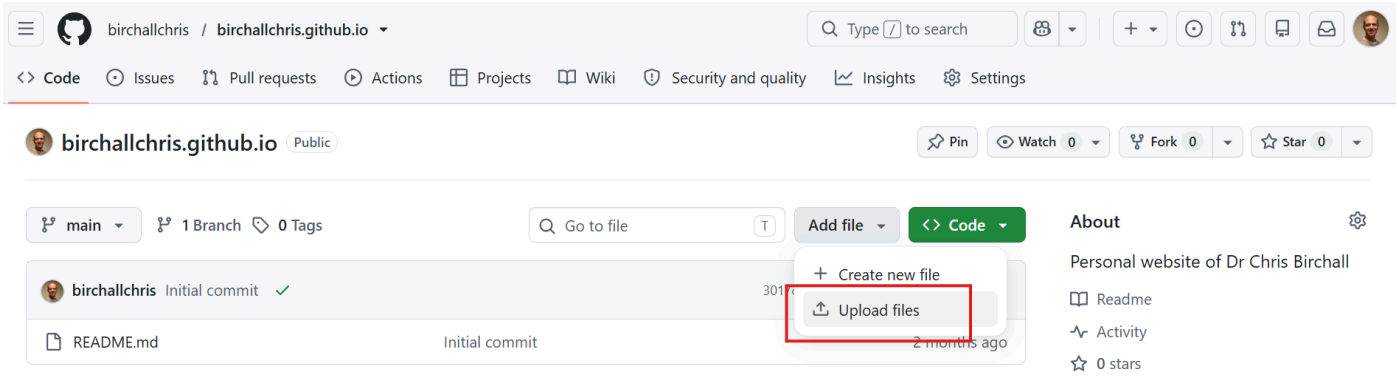
Click **Save**.

GitHub will build and publish your website using all of the files in your repository. This process may take a few minutes the first time. See below for a screenshot of settings):



You now have a website, publicly available at the URL shown in your repository Pages settings window, above the Build and deployment section (in my case https://birchallchris.github.io, as you can see in the images above).

The website will be fairly blank, as you have not uploaded your content to the repository yet. All you need to do now is upload your website files – the index.html and style.css files that you created earlier. Go back to the repository home by clicking Code in the top left of the page (main menu, not the green code button). Now, find the Add New button and click it:



Drag your web files into the resultant window, then reload your URL in a browser to see the page.

If your GitHub Pages site appears not to update after you upload new files, try opening the site in a private browsing window or perform a hard refresh (Ctrl+F5 on Windows or Cmd+Shift+R on Mac). Browsers sometimes cache older versions of GitHub Pages sites, particularly immediately after the first deployment.

GitHub Resources

GitHub is huge, powerful and complicated. We are using it for one of it's simplest features. To use GitHub fully, you should familiarise yourself with the platform and ensure that you understand how to create repositories, upload files, manage versions, and, where appropriate, publish websites using GitHub Pages. GitHub provides a range of official guides and tutorials that explain these processes step by step. Students are strongly encouraged to consult these resources before and during their project work.

Useful resources include:

- GitHub Documentation: https://docs.github.com/ (https://docs.github.com/)
- Getting Started with GitHub: https://docs.github.com/en/get-started (https://docs.github.com/en/get-started)
- Creating and Managing Repositories: https://docs.github.com/en/repositories (https://docs.github.com/en/repositories)
- GitHub Pages Documentation: https://docs.github.com/en/pages (https://docs.github.com/en/pages)
- Quickstart Guide for GitHub Pages: https://docs.github.com/en/pages/quickstart (https://docs.github.com/en/pages/quickstart)

Please note that if you choose to use GitHub, you remain responsible for maintaining copies of your work and for ensuring that all assessed materials are submitted through the University's approved assessment systems in the required format. GitHub should be viewed as a development and hosting platform rather than a substitute for formal assessment submission processes.

Validating Your Page
Visit the W3C validator website to see how your HTML code matches up to international standards: http://validator.w3.org (http://validator.w3.org)

Validation is the process of ensuring that your code meets professional standards of quality. Read the documentation at the W3 site if you are unsure what these standards are.

Forking, branching and using templates in GitHub

I have created a simple website and hosted it in GitHub using GitHub pages. Have a look at the site here: https://birchallchris.github.io/templatewebsite/ (https://birchallchris.github.io/templatewebsite/). This link takes you to a designed website, hosted just like we have seen previously, with HTML and CSS files on a web server.

Now, have a look at the page through this alternative URL: https://github.com/birchallchris/templatewebsite (https://github.com/birchallchris/templatewebsite). This is the same project, but instead of serving it through GitHub Pages, we are looking at the repository itself on GitHub.

The great thing about this is that this is open source code. You can all see, use and download that code. You can even do that officially, within the platform itself, through the use of specific sharing and collaboration tools. There are four main ways to share a project in GitHub:

- **templates** – if a project is set up as a template, other users can simply copy their own personal version of it
- **forking** – allows you to create your own version of a project in your own account, but keeps a link to the original

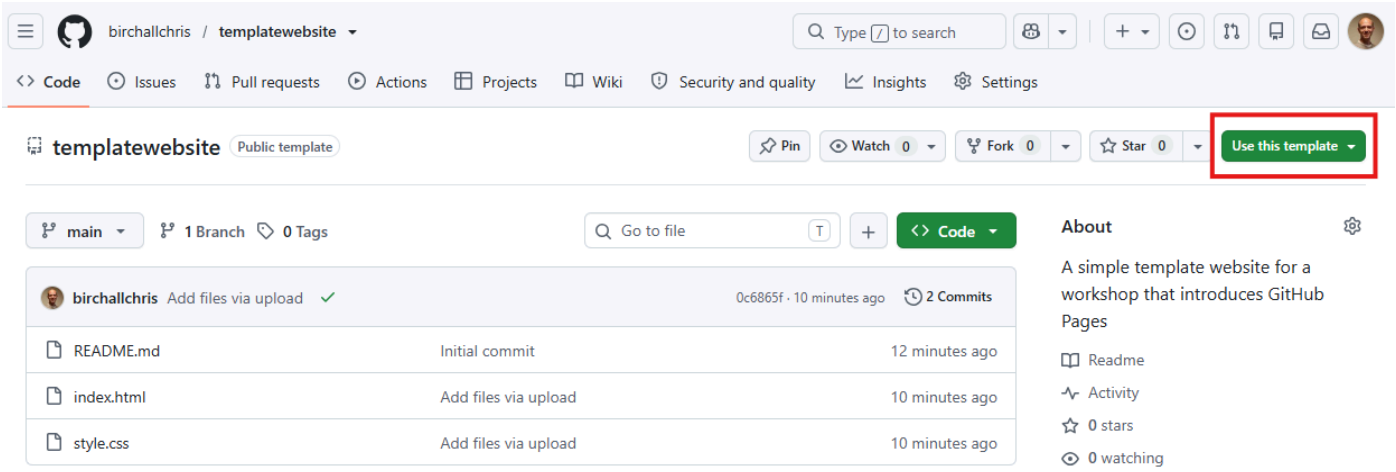
- **branching** – creates a new version of the project, within the owners account, without it becoming the live version – often used to test new code and features
- **cloning** – downloads a local version of the project to your laptop, if you are using Git software on that machine

These concepts are at the heart of open source development. Making things available to others so that they can change, adapt and improve them, instead of having to reinvent them. This approach also works really well in big teams in industry, where there are slightly different use cases for the same basic code.

The Fork button is a fundamental part of GitHub:



However, we are going to use Templates, and i have set my repository up as a template, so that you can simply copy your own version of it. Go to my repository and click the **Use This Template** button:



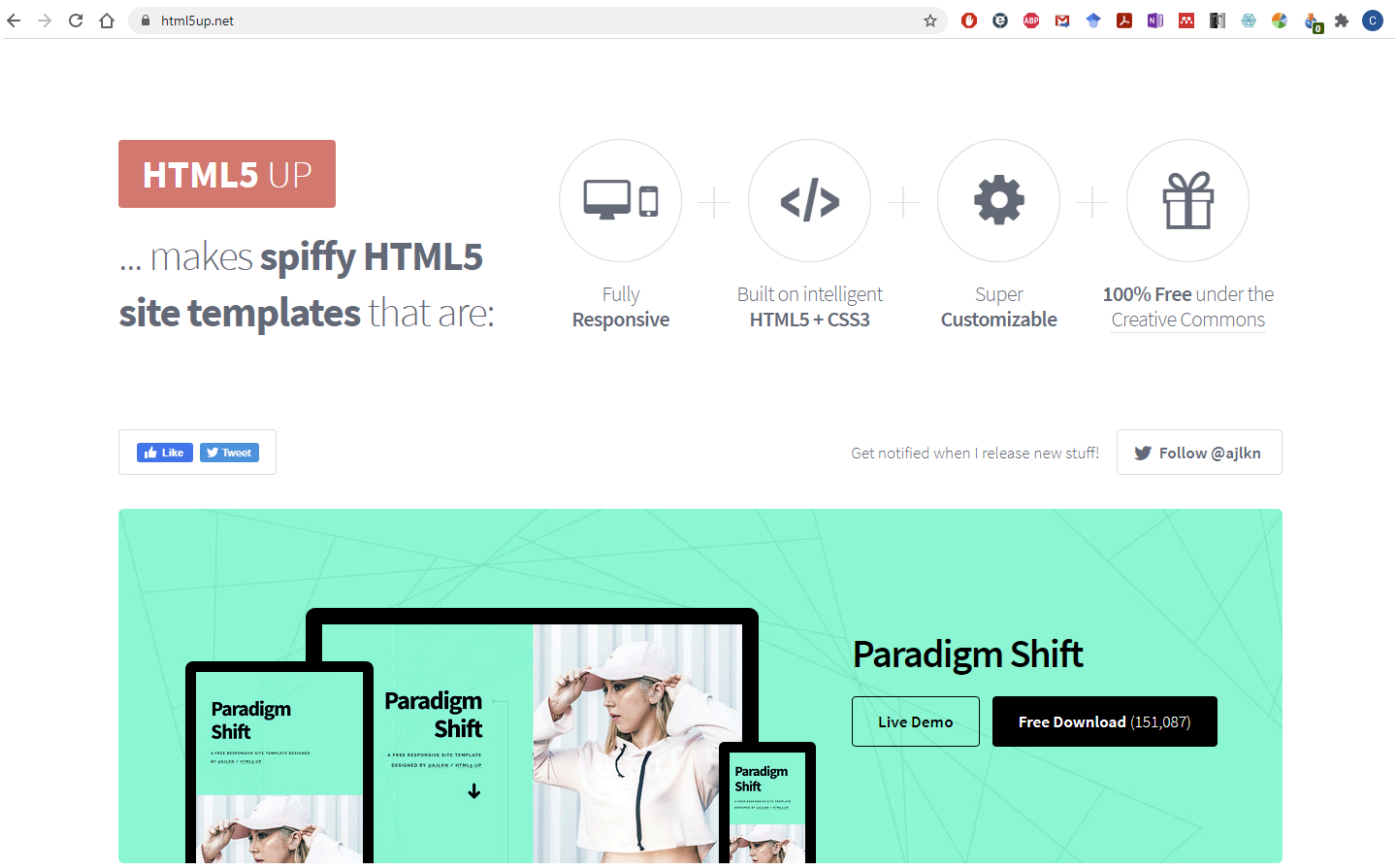
You will see that you can create your own repository or open the code itself in an online editor. Create your own repository.

Now, in your own repository, you can edit the code of the files to make your website... your website (instead of mine). You may need to download/upload, or maybe edit online. Feel free to explore the version control features in GitHub too.

Building a Portfolio

You should use one of the approaches above to showcase the work that you do on the programme. We will be expecting you to produce work to put here over the coming weeks, and your assessments in Semester 2 will be hosted online in one of these spaces. You should also try to create a website that shows you off as a digital media scholar and professional: an online portfolio of your work and your skills and experiences. You may hand code this, if you have the ability; unique, bespoke designs are often the most impressive.

However, you may want to save time and effort by using a portfolio theme in CampusPress or downloading a well-designed portfolio template from the internet and using in a GitHub Pages website. See, for example, HTML5Up.net (http://HTML5Up.net):



Simply download a template to your local machine, unzip the contents and then open and edit to add in your own content or tweak the design. Upload the whole lot into your a repository on GitHub (with Pages activated) to make your portfolio available to the world.

Sample Code

Don't know where to start? Don't worry – we've got you covered.

Basic HTML Elements

Headings:

```
<h1>The most important heading on the page</h1>
<h2>A heading at the second structural level of the document</h2>
<h3>A heading at the third structural level of the document</h3>
and so on to <h6>
```

Paragraphs:

```
<p>The first paragraph of the document. Blah blah blah.</p>
<p>The second paragraph of the document. La La La.</p>
```

Emphasis: (for italics) and (for bold)

```
<p>I did <em>not</em> mean that!</p>
<p>I <strong>really</strong> did not mean that!!</p>
```

Line breaks:

```
<p>Clothworkers' Building North<br/>
University of Leeds<br/>
Leeds LS2 9JT</p>
```

Lists:

```
<ul></ul> unordered lists
<ol></ol> ordered lists
<dl></dl> definition lists
<li></li> for list items

<ul>
<li>Roast lamb.</li>
<li>Morroccan vegetarian stew.</li>
<li>Creamy king prawn and chorizo risotto.</li>
</ul>
```

Hyperlinks:

```
<a href="newpage.html">Link to new page</a>
```

Basic CSS (Styling and Page Layout)

selector	{ property:	value; }
body	background-color:	white;
	background:	black;
	color:	gray;
h1	font-family:	Verdana,Arial,Helvetica,sans-serif;
		Georgia,"Times New Roman", Times, serif;
	margin-left:	-10%;
	margin-right:	10%;
	margin-top:	2em;
	margin-bottom:	2em;
h2	font-size:	2em;
	font-size:	140%
	text-transform:	uppercase;
	border:	solid;
p	text-indent:	2em;
blockquote	font-style:	italic;

Sample Stylesheet

```
body {
color: gray;
background: white;
font-family: Verdana,Arial,Helvetica,sans-serif;
margin-left: 10%;
margin-right: 10%;
}
h1 {
color: #457926;
font-family: Georgia,"Times New Roman",Times,serif;
font-size: 2em;
text-transform: uppercase;
margin-left: -8%;
}
h2 {
color: #457926;
font-family: Georgia,"Times New Roman",Times,serif;
font-size: 1.7em;
border: solid;
margin-left: -4%;
margin-top: 2em;
margin-bottom: 2em;
}
p {
font-size: 1.2em;
text-indent: 2em;
}
blockquote {
font-size: 1em;
font-style: italic;
color: black;
background: red;
}
.first-line {
font-weight: bold;
}
#first {
font-size: 1.4em;
color: blue;
}
```

Further Resources

Web development can be difficult, especially if you're new to it. Here are some additional resources that may be helpful.

Web Resources

- Getting started with HTML: http://www.w3.org/MarkUp/Guide (http://www.w3.org/MarkUp/Guide)
- More advanced HTML features: http://www.w3.org/MarkUp/Guide/Advanced.html (http://www.w3.org/MarkUp/Guide/Advanced.html)
- Introduction to CSS (adding a touch of style!): http://www.w3.org/MarkUp/Guide/Style.html (http://www.w3.org/MarkUp/Guide/Style.html)
- The Web Standards project: http://www.webstandards.org (http://www.webstandards.org)
- WaSP InterAct Web Standards Curriculum: http://interact.webstandards.org (http://interact.webstandards.org)
- AListApart (a website for web designers): http://www.alistapart.com (http://www.alistapart.com)
- Position Is Everything: http://www.positioniseverything.net (http://www.positioniseverything.net)
- CSS Wiki: http://css-discuss.incutio.com (http://css-discuss.incutio.com)
- SitePoint: http://www.sitepoint.com (http://www.sitepoint.com)
- BoagWorld podcast: http://boagworld.com/category/podcast (http://boagworld.com/category/podcast)
- HTML5 Doctor: http://html5doctor.com (http://html5doctor.com)
- CSS3 Click Chart: http://www.impressivewebs.com/css3-click-chart (http://www.impressivewebs.com/css3-click-chart)
- HTML5 & CSS3 Resources for Designers: http://realworldcss3.com/resources (http://realworldcss3.com/resources)
- The World Wide Web Consortium (W3C): http://www.w3.org (http://www.w3.org)
- HTML & XHTML markup: http://www.w3.org/Markup (http://www.w3.org/Markup)
- W3C homepage for CSS: http://www.w3.org/Style/CSS (http://www.w3.org/Style/CSS)
- Web Content Accessibility Guidelines: http://www.w3.org/TR/WCAG20 (http://www.w3.org/TR/WCAG20)

Interface Design Package

For those who would like to know more about the ins and outs of interface design, the following Zip folder includes materials from a module on the topic. These files cover a huge range of topics, and will be useful to anyone interested in enhancing their web development practice. These files are courtesy of Dr. Chris Birchall.

Interface Design Package (13 Chapters in a Zip File)

